

- Q.4 A full adder is characterized by (CO4)
- a) Two inputs and two outputs
 - b) Three inputs and two output
 - c) Three inputs and one output
 - d) Two inputs and one output
- Q.5 If both the inputs of J-K flip-flop are high, then it responds to (CO4)
- a) No change mode b) Set mode
 - c) Reset mode d) Toggle mode
- Q.6 The most heavily doped region in a transistor is (CO2)
- a) collector b) base
 - c) emitter d) both (a) and (c)

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The electronic device that converts ac into dc is called _____. (CO1)
- Q.8 Draw symbol of PNP transistor. (CO2)
- Q.9 Convert $(9FDA)_{16}$ to binary number. (CO3)

- Q.10 A 16:1 MUX has _____ select lines. (CO4)
- Q.11 FET has _____ input impedance. (low/high) (CO2)
- Q.12 An encoder is a _____ logic circuit.
(combinational / sequential) (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Differentiate between P type and N type semiconductors. (CO1)
- Q.14 Explain full wave bridge rectifier in brief. (CO1)
- Q.15 Write a short note on filters. (CO1)
- Q.16 Explain in brief PNP and NPN transistors. (CO2)
- Q.17 Compare JFET and BJT. (CO2)
- Q.18 Distinguish between analog and digital signal. (CO3)
- Q.19 Do the following Conversions. (CO3)
- a) $(654)_8 = (?)_{10}$
- b) $(A3B7)_{16} = (?)_8$
- Q.20 Explain half adder with the help of truth table and logical expression. (CO4)
- Q.21 Explain the working of SISO shift register. (CO4)

Q.22 Define decoder. Explain in brief 3-line to 8-line decoder. (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Describe the behavior of PN junction diode under forward and reverse biasing. Also draw its V-I characteristics. (CO1)

Q.24 Explain construction, operation and characteristics of a MOSFET. (CO2)

Q.25 Explain successive approximation A/D converter with the help of suitable diagram. (CO4)